Error Propagation in Cascaded Speech Translation under Acoustic Channel Degradations: A Black-Box Benchmark of Commercial Speech Intelligence APIs

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Abstract

Modern speech intelligence services are increasingly assembled from black-box cloud APIs, in which an automatic speech recognition (ASR) system feeds a machine translation (MT) engine in a cascade, or are replaced by end-to-end speech translation (E2E ST) models. The acoustic channels that carry speech to these systems are rarely clean, yet the sensitivity of each architectural choice to realistic channel impairments is difficult to assess because commercial systems can only be queried through their interfaces. This paper presents CascadeBench, a black-box benchmark that quantifies how eight canonical acoustic channel degradations, including multi-talker babble noise at five signal-to-noise ratios, room reverberation, telephone narrowbanding, low-bitrate coding, packet loss, amplitude clipping, and tempo distortion, propagate through the recognition and translation stages of commercial speech APIs. A stratified bilingual corpus of 480 Chinese and English sentences was synthesised with multiple neural voices, degraded in a fully controlled and reproducible pipeline, and processed by five recognition systems spanning three architectural families, together with two cascade and two end-to-end speech translation paths. Results show that large-language-model ASR services degrade several times more slowly under noise than specialised streaming models, with degradation slopes of 0.3 to 0.4 percentage points per dB against 1.7 for the specialised system in Mandarin, but a dedicated telephone-band model remains competitive only in its home band. At a signal-to-noise ratio of 0 dB the strongest cascade loses 18 percent of its

BLEU score while the weakest loses 69 percent, and end-to-end speech translation proves more fragile than a well-frontend cascade. Across twelve operating conditions, translation quality declines linearly with the recognition error rate, with a pass-through slope of 0.47 to 0.66 BLEU points per percentage point of recognition error and a correlation above 0.98, which provides system integrators with a simple engineering rule for estimating translation losses. All experiments were executed on free-tier API quotas, and the complete benchmark recipe, including the corpus, perturbation code, and per-sentence transcripts, is released for reproduction.

Keywords: speech recognition, speech translation, robustness benchmark, acoustic channel degradations, cascaded systems, cloud APIs

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1. Introduction

Speech has become a primary interface to cloud intelligence. Voice assistants, meeting transcription products, live captioning services, and multilingual communication platforms are routinely assembled from commercial automatic speech recognition (ASR) and machine translation (MT) interfaces offered by large cloud providers. In a typical deployment, audio captured by a far-field microphone or transported over a communication channel is first converted to text by an ASR service, and the resulting text is then passed to a text translation engine whenever the application crosses a language boundary. This cascade architecture dominates industrial practice because it allows each stage to be selected, replaced, and billed independently, and because the best available components at each stage are often operated by different providers. An alternative architecture, end-to-end speech translation (E2E ST), maps source-language audio directly to target-language text and has matured rapidly with large foundation models.

The acoustic pathway between a speaker and these services is rarely pristine. Speech arrives mixed with competing talkers, shaped by room reverberation, band-limited by telephone channels, compressed by low-bitrate codecs,

or corrupted by packet loss on unstable network links. The robustness of ASR systems to such degradations has been a central concern of the speech research community for decades, and controlled robustness corpora have driven substantial progress. What has changed is the identity of the systems that now dominate deployment. Commercial speech services are increasingly built on large language model (LLM) based ASR, on general-purpose multimodal foundation models that accept audio alongside text, and on streaming speech translation models, none of which can be fine-tuned, inspected, or benchmarked offline by their consumers. A systems integrator who composes such services into a translation pipeline has no principled way to predict how recognition errors induced by an acoustic channel will propagate into the final translation, or whether a cascade or an end-to-end design is the safer choice under the conditions of a given deployment.

The evaluation gap is not academic. Deployment decisions are made weekly by integrators who must choose among service tiers and architectural patterns with nothing but clean-audio demo pages and marketing claims, while the operating conditions of real audio pathways, conference rooms, vehicles, telephone trunks, and congested wireless links, are precisely the conditions that clean demos do not exercise. A deployment that selects a recognition frontend on clean-audio accuracy alone can be quietly losing most of its translation quality in the field, and a deployment that assumes end-to-end robustness can be more fragile than the cascade it replaced. Closing this gap requires measurement that is controlled at the acoustic layer, complete at the pipeline layer, and reproducible at the service layer, and that is the design brief of the benchmark presented here.

This gap motivates the present study. Academic robustness benchmarks such as Aurora [1] and the CHiME challenge series [2] evaluate open recognizers on fixed noise corpora, but they do not cover the black-box commercial systems that now carry production traffic, they rarely extend beyond recognition to the downstream translation stage, and they cannot be re-run on new model versions without access to model internals. Conversely, studies that compare commercial

ASR services report error rates on fixed test sets but typically consider clean audio and stop at the transcription boundary. The propagation of acoustic degradation through a complete speech translation pipeline, comparing architectural families and both cascade and end-to-end designs under identical and fully controlled channel conditions, has to our knowledge not been quantified for the current generation of speech APIs.

This paper presents CascadeBench, a black-box benchmark designed to answer four questions. First, how sensitive are the main architectural families of commercial ASR, namely specialised streaming models, LLM-based recognisers, and general-purpose multimodal models, to eight canonical acoustic channel degradations? Second, how does recognition accuracy scale with noise severity, and do the families differ in their degradation slopes? Third, when recognition errors flow into a downstream translation engine, does the cascade amplify them, and how does the cascade compare with end-to-end speech translation under identical conditions? Fourth, what are the latency consequences of each architectural choice? To answer these questions under the constraint that only public interfaces are available, the benchmark synthesises a stratified bilingual corpus of 480 Mandarin Chinese and English sentences with multiple neural voices, applies a grid of eight controlled acoustic degradations together with a five-level noise severity sweep, and measures five recognition systems, two cascade translation paths, and two end-to-end speech translation paths through their public APIs, using only free service quotas.

The study yields findings that are relevant to both the science of robust speech processing and the practice of assembling speech services. Recognition services from the three families differ sharply and systematically in their sensitivity to noise: the specialised streaming recogniser degrades at about 1.7 percentage points per dB of noise in Mandarin, whereas the LLM-based recognisers degrade at 0.3 to 0.4 points per dB, and at a signal-to-noise ratio of 0 dB the gap between families reaches a factor of two to three in error rate. The end-to-end translation paths, often assumed to be more robust than cascades because they avoid error propagation, in fact degrade more strongly under severe noise

than a cascade built on a robust recognition frontend. Across twelve operating conditions, the BLEU score of a cascade falls linearly with its recognition error rate, with a pass-through slope of 0.47 to 0.66 BLEU points per percentage point of recognition error and a correlation coefficient above 0.98, which gives integrators a simple and accurate engineering rule for estimating translation losses from recognition quality alone. Finally, the benchmark surfaces behavioural differences that are invisible in clean-audio evaluations, including the refusal of one recognition service to return any transcript for a fraction of severely noised utterances and the collapse of a telephone-band specialist outside its home bandwidth.

The contribution of this paper is threefold. It provides, to our knowledge, the first systematic measurement of acoustic degradation propagation across recognition and translation stages for commercial black-box speech APIs, covering three architectural families of recognisers and both cascade and end-to-end translation designs. It introduces a fully reproducible benchmark recipe in which the test material, the degradations, and every model response are controlled, logged, and releasable, so that the measurements can be repeated as providers update their models. And it distils the measurements into quantitative design guidance, including degradation slopes per architectural family, the error pass-through rule for cascades, and a dual-reference protocol that exposes and controls the systematic bias that arises when automatic translation quality is scored against references produced by models from the same family as one of the systems under test.

The remainder of the paper is organised as follows. Section 2 reviews related work on robust ASR, cascade and end-to-end speech translation, and the evaluation of commercial services. Section 3 describes the benchmark design, including the corpus, the degradation grid, the systems under test, and the evaluation protocol. Section 4 presents the results, and Section 5 discusses their implications for practitioners and for future evaluation practice. Limitations are examined in Section 6, and Section 7 concludes.

2. Related work

2.1. Robustness of speech recognition under acoustic degradation

The sensitivity of ASR to noise and channel distortions has been studied since the earliest statistical recognisers [3]. Surveys such as that of Virtanen et al. [4] organise this literature around feature enhancement, model adaptation, and multi-condition training. Controlled evaluation infrastructure has been decisive for progress. The Aurora framework [1] standardised noise types and signal-to-noise ratios for telephone-band recognition, and the CHiME challenge series [2] extended such evaluation to far-field, reverberant, and overlapping-speech conditions. Classical feature transformations such as RASTA filtering [5] targeted the same distortions from the signal-processing side, and synthetic room impulse responses generated with the image method of Allen and Berkley [6] and its modern implementations [7] remain standard tools for controlled reverberation studies. More recently, data augmentation strategies such as SpecAugment [8] have embedded artificial degradation into the training process itself, and weakly supervised training at scale, as exemplified by Whisper [9], has produced recognisers with markedly improved noise tolerance. The present work inherits the methodological principle of these efforts, namely controlled and reproducible degradation of the input signal, but applies it to black-box services rather than to trainable models, which precludes training-based countermeasures and makes the measurement itself the object of study.

2.2. Cascaded and end-to-end speech translation

The cascade versus end-to-end question has accompanied speech translation since neural systems became practical. Early sequence-to-sequence systems demonstrated that audio-to-text translation without an explicit recognition stage is feasible [10, 11], and Bentivogli et al. [12] compared the two paradigms systematically on IWSLT data, finding that end-to-end systems could match cascades in translation quality while differing in error structure, with cascades inheriting recognition errors and end-to-end systems producing more

fluent but occasionally freer output. The survey of Sperber and Paulik [13] synthesises this literature, including work on robustness of end-to-end systems to noisy input. Subsequent research addressed noise robustness of end-to-end ST through self-training and data selection [14]. These studies evaluate models that the researchers train themselves on public corpora. Robustness of the end-to-end component in particular has been treated as a training-data problem, with noise-augmented source audio, self-training on noisy transcripts, and domain adaptation as the standard remedies. That framing presumes control over the model, which is exactly what a service consumer lacks. When the translation component is a fixed API, the robustness question shifts from how to train the model to how to select among available models and architectures under measured channel conditions, and that selection question is what the present benchmark operationalises. The commercially relevant question, namely how the two architectures behave when every stage is a fixed, provider-operated service selected and combined by an integrator, has received far less attention, and the present benchmark is designed specifically for that setting.

2.3. Evaluation of commercial speech services

A third strand of related work evaluates commercial speech services as they are exposed to users. Such studies necessarily operate through public APIs, report error rates on fixed test material, and periodically re-run as services evolve. Their findings consistently show substantial differences between providers and between languages. Methodologically, these evaluations face constraints that mirror ours: no access to model internals, no control over model versions, and reliance on text normalisation to compare outputs against references. They have, however, largely concentrated on clean or moderately degraded audio, and on transcription accuracy alone. The broader evaluation culture in speech and audio quality research, with its calibrated degradation methodology and emphasis on reproducibility, similarly targets the acoustic layer rather than the linguistic stages downstream, and it is that tradition which the present service-level benchmark follows. The contribution of the present study relative to this

strand is twofold. First, the evaluation is extended across the translation boundary, quantifying how recognition differences propagate into translation quality for both cascade and end-to-end architectures. Second, the acoustic degradations are applied under full experimental control, with calibrated noise levels, a synthetic room impulse response, codec and packet-loss emulation, and fixed random seeds, which makes the entire measurement reproducible rather than dependent on whatever happened to be in the test recordings.

2.4. Positioning

In summary, robustness benchmarks address open models but not commercial services, service evaluations address commercial ASR but usually only in clean conditions and only at the recognition boundary, and the cascade-versus-end-to-end literature addresses research models rather than provider-operated systems. CascadeBench combines the three perspectives. It applies the controlled degradation methodology of the robustness literature to the black-box commercial setting, extends the measurement through the translation stage for both architectures, and adds a dual-reference scoring protocol that quantifies a bias source that any evaluation involving modern generative reference translations must confront. To our knowledge, no previous study has measured all of these dimensions simultaneously for the current generation of speech APIs, and none has reported the linear recognition-to-translation error pass-through relationship that Section 4 documents.

3. Materials and methods

3.1. Design overview

CascadeBench measures how acoustic channel degradations propagate through commercial speech recognition and translation services that can only be accessed through their public interfaces. The design is shown in Fig. 1. A bilingual corpus of 480 sentences is synthesised to audio with multiple neural voices, degraded by a grid of controlled channel impairments, and submitted to five recognition systems and four speech translation paths. Every stage of the pipeline is scripted

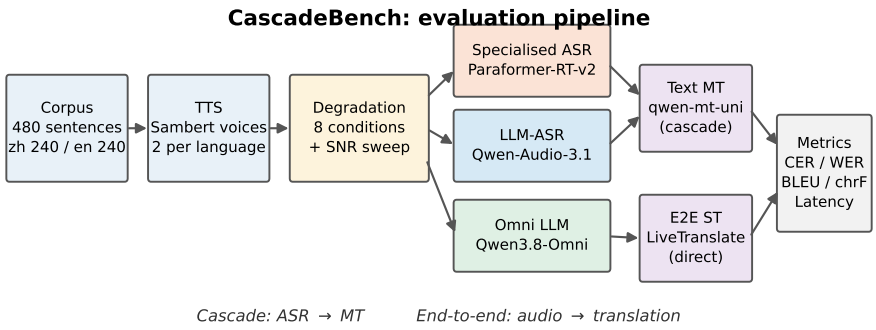


Figure 1: Architecture of the CascadeBench evaluation pipeline. A stratified bilingual corpus is synthesised with multiple neural voices, degraded by a grid of eight acoustic channel conditions plus a noise severity sweep, and processed by recognition systems from three architectural families. Cascade systems forward recogniser output to a text translation engine; end-to-end systems translate audio directly.

with fixed random seeds, and every model response together with its round-trip latency is logged at the sentence level, so that the complete measurement can be re-executed as providers update their services.

3.2. Corpus

The corpus contains 240 Mandarin Chinese and 240 English sentences, balanced across four content domains and three length strata, twenty sentences per cell. The domains are broadcast news, everyday conversation, technical and scientific exposition, and a numerals-and-entities stratum rich in dates, quantities, addresses, and named entities, which is known to stress recognition and inverse text normalisation. Short sentences contain 8 to 14 Chinese characters or 5 to 9 English words, medium sentences 15 to 25 characters or 10 to 17 words, and long sentences 26 to 39 characters or 18 to 24 words. The mean length is 19.3 characters for Chinese and 12.0 words for English. Sentences were authored to be self-contained and factually neutral, to avoid brand names and

transliterated loanwords whose recognition would depend on provider-specific vocabulary rather than on acoustic processing, and to distribute numerical expressions across date, time, quantity, price, and identifier formats within the numerals domain. All numbers are written out in words in the corpus text, for example two thousand twenty five, so that the spoken realisation is unambiguous and the reference transcript matches the synthesis input exactly.

Each sentence was synthesised by a neural speech synthesis service (Sambert, Alibaba Cloud Model Studio) using two distinct voices per language, a male and a female voice for Chinese and two native-sounding voices for English, with sentences alternating between the voices so that each voice realises 120 sentences per language. Synthesis produced 16 kHz single-channel PCM audio. The 44 babble noise source tracks described below were synthesised with six additional voices that are disjoint from the four test voices. Using synthesis rather than recorded speech has two methodological advantages. The reference transcript is exact by construction, which removes the transcription noise that limits error floor measurement in recorded corpora, and the acoustic properties of the test material can be regenerated at any time, which decouples the benchmark from the licensing and availability constraints of recorded corpora such as AISHELL-1 [15] or LibriSpeech-derived material [16]. The corresponding limitation, that synthetic speech does not sample the full variability of spontaneous human speech, is examined in Section 6.

3.3. Acoustic degradation grid

Eight channel conditions plus the clean control form the main grid. The conditions were selected to span the canonical impairment families that speech services encounter between a talker and a data centre.

Additive babble noise is the most extensively sampled family. Babble was constructed by concatenating randomly selected segments from the 44 disjoint noise-source tracks described above, mixing three such streams, and normalising the mixture, which produces speech-shaped noise with realistic spectral and temporal structure rather than stationary synthetic noise. Speech-shaped inter-

ference was chosen over stationary noise because it is the harder and the more ecologically valid case: babble exhibits the temporal modulation and spectral occupancy of competing talkers, masking the target in the same time-frequency regions that carry linguistic information, whereas stationary noise is partly countered by any spectral-subtraction-style front-end that a provider may have deployed internally. The noise was scaled against the active-speech level of each utterance, measured as the mean energy of frames above the 60th energy percentile, to hit target signal-to-noise ratios of 20, 15, 10, 5, and 0 dB. Calibration was verified by re-measuring the realised signal-to-noise ratio of the mixed signals against the clean signal; the realised values matched the targets within 0.1 dB on average. The 10 dB and 0 dB points enter the main grid, and the full five-point ladder forms the severity sweep on a stratified 144-sentence subset, 72 sentences per language, sampled evenly from every domain-by-length cell.

Reverberation was applied by convolving each signal with a synthetic room impulse response of the exponentially decaying noise form [6, 7] with a direct-path impulse and a reverberation time of 0.5 s, which corresponds to an ordinary reverberant room rather than an extreme auditorium. Narrowband degradation resamples each signal to 8 kHz and back to 16 kHz, which removes the 4 to 8 kHz band exactly as a telephone channel does; the telephone-specialised recogniser receives the 8 kHz signal natively. Codec degradation encodes each signal with the Opus codec [17] at 8 kbit/s mono and decodes it back, emulating an aggressive but deployed voice-over-IP setting. Packet loss zeroes 15 percent of 20 ms frames chosen uniformly at random, emulating lossy transport without jitter compensation. Clipping applies hard amplitude limiting at 0.3 of full scale. Tempo distortion resamples the time axis by a factor of 1.15 with pitch preservation, emulating playback-speed variation or time-scale modification upstream of the service. All random draws use per-sentence and per-condition seeds derived deterministically from the sentence identifier and condition name, so the degraded signals are bitwise reproducible.

Formally, for a clean signal $x[n]$ of length N and a babble realisation $b[n]$,

the mixed signal at level ρ dB is

$$y[n] = x[n] + \sigma\, b[n], \qquad \sigma = \sqrt{\frac{P_{\mathrm{act}}(x)}{10^{\rho/10}\,\overline{P}(b)}}, \tag{1}$$

where $\overline{P}(b)$ is the mean power of the babble and $P_{\mathrm{act}}(x)$ is the active-speech power estimated from frames whose energy exceeds the 60th energy percentile of the utterance, which approximates the ITU-T P.56 active-level convention with a fixed percentile rule. The reverberation impulse response is

$$h[n] = \delta[n] + v[n]\, e^{-6.9\, n/(f_s T_{60})}, \qquad n = 0, \ldots, \lceil f_s T_{60} \rceil, \tag{2}$$

with $v[n]$ unit-variance white noise, $f_s = 16$ kHz, $T_{60} = 0.5$ s, and h normalised to unit energy, and the reverberant signal is the truncated convolution of x with h rescaled to the peak amplitude of x. Packet loss zeroes each independent 20 ms frame with probability $p = 0.15$,

$$y[n] = m_k\, x[n], \quad n \in \text{frame } k, \qquad m_k \sim \text{Bernoulli}(1-p), \tag{3}$$

and clipping applies $y[n] = \mathrm{sign}(x[n])\min(|x[n]|, 0.3)$. Narrowbanding resamples by factors of one half and two with polyphase anti-aliasing filters, codec degradation round-trips the signal through the Opus encoder at 8 kbit/s, and tempo distortion applies a pitch-preserving time-scale modification of factor 1.15.

3.4. Systems under test

Table 1 lists the systems. The recognition systems span three architectural families as served by a single commercial provider (Alibaba Cloud Model Studio, Beijing region) during October 2026, accessed through their documented interfaces. The specialised family consists of streaming recognisers in the Paraformer lineage [18], a 16 kHz model and an 8 kHz telephone-band model. The LLM-ASR family consists of Qwen-Audio 3.1 Flash [19] and Qwen3-ASR Flash, audio input models built on large language model backbones. The multimodal family consists of Qwen3.8-Omni Flash, a general-purpose omni-modal foundation model [20] queried with a transcription instruction. All systems were accessed in their default configurations, since a consumer of a black-box service cannot

apply front-end training or adaptation, and the question of interest is precisely how the services behave out of the box. The streaming recognisers were driven by streaming local audio over their realtime WebSocket interface, sending each utterance as 16 kHz or 8 kHz PCM frames at the maximum accepted rate and collecting sentence-final segments. The LLM-ASR and omni systems received base64-encoded whole utterances in their audio-input message format, the omni model together with an instruction prompt for transcription or translation. The streaming translation service was driven over its realtime interface with a session configuration specifying text-only output and the target language, sending raw PCM frames and collecting both the source transcript and the translated text that the service emits, and the text translation engine received source text with explicit source and target language tags. Every call was executed with an automatic retry policy of up to six attempts with exponential backoff on throttling, timeout, and connection failures, and no other tuning parameters were exposed or set. The speech translation systems are the text translation engine qwen-mt-uni, used as the second stage of two cascades built on the 16 kHz specialised recogniser and on the Qwen-Audio 3.1 recogniser, and two end-to-end paths: the omni foundation model with an audio translation instruction, and a dedicated streaming speech translation service (LiveTranslate) that internally combines a streaming frontend recogniser with a translation model. The internal frontend of the streaming translation service, which its responses expose as source transcripts, is additionally scored as a sixth recognition system. All API calls were executed on free-tier quotas with the stop-on-exhaustion option enabled, which both bounds the study at zero monetary cost and documents the practical quota requirements of such a measurement.

3.5. Evaluation protocol

Recognition accuracy is scored as the character error rate (CER) for Chinese and the word error rate (WER) for English, computed with the standard Levenshtein alignment [21] after a normalisation pipeline that is applied identically to references and hypotheses. The normalisation performs Unicode compatibility

Table 1: Systems under test and their roles in the benchmark.

System (family)	Interface	Role	Evaluated on
Paraformer-RT-v2 (specialised)	streaming ASR	recogniser	full grid
Paraformer-RT-8K-v2 (specialised, 8 kHz)	streaming ASR	recogniser	Chinese subset
Qwen-Audio-3.1-Flash (LLM-ASR)	audio input API	recogniser	full grid
Qwen3-ASR-Flash (LLM-ASR)	audio input API	recogniser	full grid
Qwen3.8-Omni-Flash (omni)	audio input API	recogniser, E2E ST	main grid
LiveTranslate frontend (specialised)	streaming ST	recogniser (internal)	sweep subset
qwen-mt-uni (MT)	text translation	cascade stage, oracle	full corpus
Qwen3.8-Max (LLM)	text generation	independent reference	full corpus

folding, case folding, punctuation removal, and, critically, canonicalisation of numeric expressions. Because commercial recognisers differ in their inverse text normalisation conventions, with the same spoken quantity rendered as twenty six, 26, or two six depending on the system, both sides of every comparison are mapped to a canonical digit form by rules that handle Chinese and English cardinal numbers, digit-by-digit readings, time expressions, decimal fractions, and scale multipliers such as wan and yi on the Chinese side and million and billion on the English side, with ordinal expressions masked to a common token. The rule set was developed and verified on the clean condition, where residual normalisation artefacts were inspected sentence by sentence until the clean-condition error rates reported in Section 4 reflect genuine recognition behaviour rather than scoring artefacts. Utterances for which a service returns no transcript at all, a behaviour documented in Section 4, are scored as complete failures with an error rate of one.

Translation quality is scored with corpus-level BLEU [22] using sacreBLEU [23] with the 13a tokeniser for English targets and the character-level tokeniser for Chinese targets, and with chrF [24] as a second string-matching metric. Because the corpus is direction-symmetric, BLEU is computed separately for the Chinese-to-English and English-to-Chinese halves and macro-averaged. Every hypothesis is scored against two reference sets. The primary reference is pro-

duced by Qwen3.8-Max [25], a general-purpose LLM from a different product family than any translation system under test, translating the clean reference transcripts. The secondary reference is the oracle cascade output, that is, the cascade translation engine applied to the clean reference transcripts. The dual-reference design exposes a bias that Section 4.5 quantifies: automatic scoring against machine-generated references systematically favours systems from the same model family as the reference generator, and conclusions about absolute quality depend on the choice of reference, whereas conclusions about degradation are stable.

Latency is measured as the wall-clock round trip of each API call from dispatch to the complete response, recorded per call. For cascades, the reported end-to-end latency is the sum of the recognition and translation call latencies, which is the quantity a service consumer experiences when the two stages are invoked sequentially. Streaming systems are fed audio at the maximum rate their interface accepts, so the measurement reflects service-side processing rather than client-side throttling.

3.6. Experimental scale and cost

The complete measurement comprises 26,238 successful API calls: 520 synthesis calls producing 31 minutes of source audio, 4,752 degraded variants per recognition system covering 4.9 hours of audio, and 7,411 translation calls across the oracle, reference, cascade, and end-to-end phases. Another 3,400 calls were consumed by retry attempts and by connection resets of the realtime translation interface, which are counted here because they represent real operational behaviour of the service. The entire campaign ran on complimentary service quotas with the stop-on-exhaustion option enabled, and one system, the omni translation path, reached its free-tier limit before completing the severity sweep, which bounds the end-to-end comparison to the main grid with 84 of the 144 sweep sentences per condition. The total wall-clock duration of data collection was approximately four hours with up to five systems queried in parallel. These figures are reported so that other laboratories can estimate the feasibility of

replicating or extending the measurement; they also demonstrate that a full multi-system, multi-condition robustness characterisation of commercial speech services is attainable at zero monetary cost with careful quota budgeting, which lowers the barrier for the periodic re-measurement that the field currently lacks.

3.7. Statistical analysis

All aggregate error rates and quality scores are reported with 95 percent bootstrap confidence intervals over sentences, using 1000 resamples. Differences between systems within a condition are tested with the Wilcoxon signed-rank test [26] on paired per-sentence errors, with Holm correction [27] across the family of pairwise comparisons. Noise severity is modelled by ordinary least squares regression of the error rate on the signal-to-noise ratio, with bootstrap confidence intervals for the slope, and the slope is reported in percentage points per dB as the degradation gradient of a system. The relationship between recognition error and translation quality across operating conditions is summarised by the cascade amplification coefficient, defined as the slope of regressing the BLEU loss from the oracle against the mean recognition error rate, together with its Pearson correlation; a slope near one with high correlation indicates linear error pass-through, and the slope value itself is the pass-through rate. Signal processing and statistics use SciPy [28] and librosa [29].

4. Results and analysis

4.1. Clean baseline and the main degradation grid

Table 2 reports sentence-level error rates on the main grid for the five recognition systems, and Fig. 2 visualises the same data split by language. On clean synthetic speech the three foundation-model recognisers are close to their scoring floor, with error rates between 0.3 and 0.5 percent, while the specialised streaming recogniser operates at 0.56 percent on Chinese and 3.99 percent on English, reflecting its primarily Mandarin training. The telephone-band specialist reaches 1.86 percent on its native language. The clean-condition figures

Table 2: Mean recognition error rates in percent on the main degradation grid, both languages pooled, with bootstrap 95 percent confidence intervals in parentheses. The 8 kHz specialist is evaluated on the Chinese subset only.

System	Clean	Babble 10	Babble 0	Reverb	Narrow	Codec	Loss	Clip	Tempo
Paraformer-RT-v2	2.3	2.8	58.5	3.2	2.1	2.4	3.2	2.5	2.3
	(1.7,2.9)	(2.2,3.6)	(54.5,62.6)	(2.5,4.0)	(1.5,2.7)	(1.8,3.1)	(2.5,4.0)	(1.9,3.2)	(1.7,3.0)
Paraformer-RT-8K	1.9	4.7	73.2	18.2	1.9	1.8	3.3	1.9	2.3
	(1.2,2.7)	(3.3,6.3)	(68.8,77.6)	(14.4,22.3)	(1.2,2.7)	(1.1,2.6)	(2.2,4.6)	(1.2,2.7)	(1.5,3.3)
Qwen-Audio-3.1	0.5	0.9	28.2	0.7	0.6	0.6	0.6	0.9	0.6
	(0.3,0.7)	(0.6,1.3)	(24.8,32.0)	(0.5,1.0)	(0.4,0.8)	(0.4,0.8)	(0.4,0.9)	(0.6,1.2)	(0.4,0.8)
Qwen3-ASR-Flash	0.3	0.5	25.1	0.3	0.3	0.3	0.4	0.3	0.3
	(0.2,0.5)	(0.3,0.8)	(21.7,28.7)	(0.2,0.5)	(0.2,0.5)	(0.2,0.5)	(0.2,0.6)	(0.2,0.5)	(0.2,0.5)
Qwen3.8-Omni-Flash	0.4	0.6	31.8	1.1	0.1	0.3	0.2	1.0	0.3
	(0.2,0.6)	(0.4,0.9)	(28.0,35.8)	(0.7,1.6)	(0.0,0.3)	(0.1,0.5)	(0.1,0.4)	(0.6,1.5)	(0.1,0.5)

establish that the comparisons under degradation are not confounded by differences in clean accuracy of comparable magnitude.

Three regularities stand out in the grid. First, babble noise at 0 dB is by far the most destructive condition for every system, but the magnitude of the collapse separates the architectural families sharply. The specialised streaming recogniser reaches a pooled error rate of 58.5 percent and the telephone-band specialist 73.2 percent, whereas the two LLM-ASR systems remain at 25 to 28 percent and the omni model at 31.8 percent. At matched severity, the foundation-model recognisers commit roughly half the error of the specialised systems, and the difference is significant at the sentence level: in Chinese at 0 dB, Qwen-Audio 3.1 errs at 6.3 percent against 41.6 percent for the streaming specialist (Wilcoxon $p < 0.001$, $n = 240$).

Second, the effect of noise is strongly language-asymmetric in a way that differs by family. For the LLM-ASR systems, Chinese recognition at 0 dB is nearly immune, at 6.3 percent for Qwen-Audio 3.1 and 13.1 percent for Qwen3 ASR, while their English error rates at the same severity reach 50.2 and 37.1

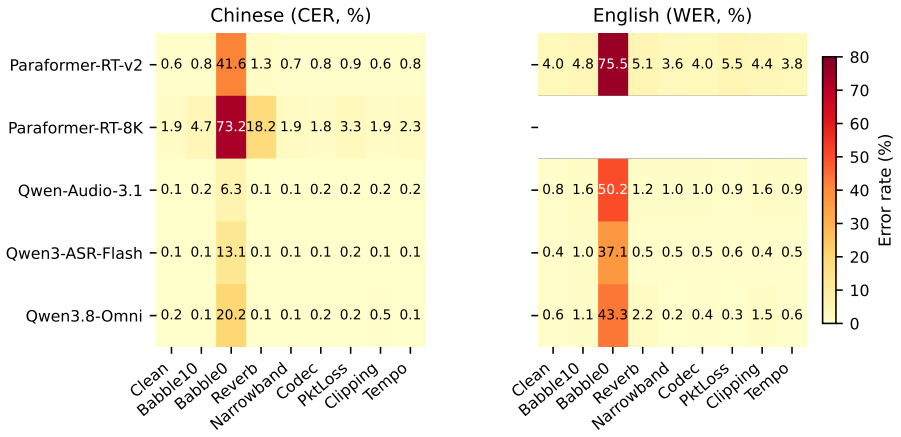


Figure 2: Recognition error rates by system and degradation condition, separated by language. Colour scale is truncated at 80 percent. The 8 kHz specialist is evaluated on the Chinese subset only.

percent. The specialised streaming recogniser suffers heavily in both languages, 41.6 percent in Chinese and 75.5 percent in English. The babble mixture is language-mixed, so the asymmetry cannot be attributed to language-matched interference alone; a plausible partial explanation is that the tonal structure of Mandarin provides complementary cues that survive energetic masking, a hypothesis consistent with the general observation that Mandarin recognition degrades more slowly than English under stationary interference [4].

Third, the non-noise degradations expose narrow specialisation rather than uniform fragility. Narrowbanding to the telephone band costs the LLM-ASR systems almost nothing, 0.3 to 0.6 percent, and the omni model is essentially unaffected at 0.1 percent, whereas the telephone-band specialist shows no measurable home-field advantage over the generalist systems in its own band. Reverberation, by contrast, is disproportionately expensive for the telephone-band specialist at 18.2 percent, an order of magnitude above every other system, which suggests that its front-end processing is tuned for direct-path telephone speech. Codec compression, packet loss, clipping, and tempo distortion at the chosen severities are individually mild for all systems, raising error rates by at

most one percentage point over clean speech. This does not make them harmless: Section 4.4 shows that even sub-point recognition changes propagate measurably into translation quality, and the conditions were deliberately calibrated to deployed-but-aggressive levels rather than to failure-inducing extremes.

Two further behaviours deserve mention because they are invisible in conventional aggregate reporting. The numeral-and-entity domain is five times more error-prone than the other domains for Qwen-Audio 3.1 on clean speech, at 1.25 against 0.25 percent, whereas Qwen3 ASR shows no such penalty, which indicates that inverse text normalisation rather than acoustics dominates that gap. More strikingly, Qwen-Audio 3.1 is the only system that ever declines to answer: on 5.2 percent of the 0 dB utterances, concentrated in English at 9.6 percent of that subset, it returns an empty transcript, and these refusals are scored as complete failures in Table 2. The other four systems always emit a hypothesis, however wrong. For a downstream consumer this is an operationally meaningful difference between a wrong answer and no answer, and it is the kind of behavioural divergence that only black-box evaluation at controlled severity can surface.

The remaining conditions merit condition-specific remarks because their effects are small in aggregate but structured in detail. Reverberation with a reverberation time of 0.5 s raises the specialised streaming recogniser to 3.2 percent pooled, but the effect concentrates in English, where it reaches 5.1 percent against 1.3 percent in Chinese, and the paired difference against the LLM-ASR systems remains significant in both languages (Wilcoxon $p < 0.01$ in Chinese, $p < 0.001$ in English). The omni model shows an idiosyncratic signature under reverberation and clipping, at 1.1 and 1.0 percent, roughly triple its clean error rate, which suggests that its audio encoder is more sensitive to spectral colouration and non-linear distortion than the dedicated recognisers even though it dominates under noise. Codec compression at 8 kbit/s leaves every system within 0.3 percentage points of its clean score, confirming that deployed neural codecs are essentially transparent to modern recognisers at this bitrate. Packet loss at 15 percent and clipping at the 0.3 threshold behave as mild aggregate

stressors, and tempo distortion at 1.15 times is fully transparent to every system, which is expected because time-scale modification preserves the spectral envelope that recognisers rely on. The practical reading of these patterns is that a service integrator who cannot control the acoustic environment should prioritise noise mitigation far above codec or transport tuning, since the latter contribute almost nothing to the error budget at realistic operating points.

4.2. Stratified analysis by domain

Because the corpus is balanced by construction, error rates can be decomposed by content domain without re-weighting. On clean audio the technical domain is the most error-prone for the specialised streaming recogniser at 3.3 percent, consistent with its terminological density, while the conversation domain is the easiest for every system, below 0.1 percent for the Qwen-Audio recogniser. The numerals-and-entities domain shows the clearest family split on clean speech: the two LLM-ASR systems differ by an order of magnitude on it, at 1.2 percent for Qwen-Audio against 0.15 percent for Qwen3-ASR, which as noted in Section 4.1 points to inverse text normalisation rather than acoustics as the separating factor. Under 0 dB noise the domain differences compress for every system: the spread across domains narrows to about nine points for the Qwen-Audio recogniser and to twelve points for the specialised streaming recogniser, and the ordering changes, with the technical domain becoming the most difficult for both. The compression indicates that under severe energetic masking the bottleneck shifts from lexical and normalisation knowledge, which distinguishes systems on clean speech, to the acoustic recovery problem, which is common to all systems and dominates the error budget.

4.3. Scaling with noise severity

Fig. 3 shows error rate as a function of signal-to-noise ratio on the 144-sentence severity sweep, and Table 3 reports the fitted degradation gradients. The relation is approximately linear over the measured 20 to 0 dB range for all systems, which makes the slope a compact summary of noise robustness. The

Table 3: Noise degradation gradients, in percentage points of error rate per dB of signal-to-noise ratio, fitted over the 20 to 0 dB sweep with bootstrap 95 percent confidence intervals.

System	Chinese	English
Paraformer-RT-v2	-1.70 $(-2.05, -1.36)$	-3.08 $(-3.72, -2.49)$
Qwen-Audio-3.1	-0.23 $(-0.33, -0.14)$	-1.30 $(-1.66, -0.96)$
Qwen3-ASR-Flash	-0.44 $(-0.60, -0.30)$	-1.35 $(-1.75, -0.97)$

specialised streaming recogniser loses 1.70 percentage points of Chinese accuracy per dB and 3.08 points per dB in English, whereas the LLM-ASR systems lose 0.34 to 0.44 points per dB in Chinese and 1.35 to 1.57 in English. In relative terms the specialised system degrades four to five times faster in Chinese and about twice as fast in English. The omni model, measured on the main grid points only, tracks the LLM-ASR family. The crossover structure visible in Fig. 3 carries the practical content: above roughly 5 dB the systems are within a point of each other and choice can be driven by latency or cost, but below 5 dB the families separate rapidly, and below 0 dB no system delivers usable English transcription, with error rates between 37 and 76 percent. The sweep also shows that the separation is gradual rather than threshold-like. Between 20 and 10 dB every system stays within one percentage point of its clean score, so the mild-noise regime is effectively free of robustness concerns for all families. Between 10 and 5 dB the specialised system begins to move first, and by 5 dB the family ordering is already established in both languages, which means that robustness differences can be detected at moderate severity without stressing any system to failure. The practical implication for procurement testing is that a single 5 dB babble probe suffices to rank candidate recognition services by noise robustness in this measurement, since the ordering at 5 dB matches the ordering at 0 dB in both languages for every pair of systems compared.

4.4. Cascade versus end-to-end translation under degradation

Fig. 4 compares the four speech translation paths across the main grid, and Table 4 reports the corresponding scores. On clean audio the two cascades lose

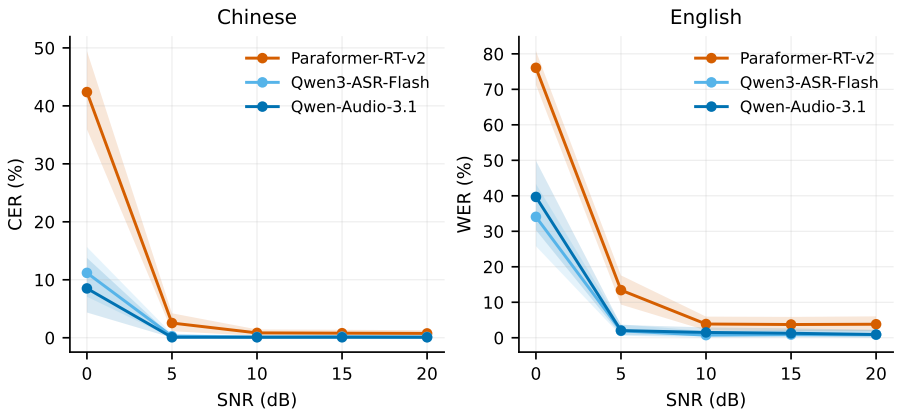


Figure 3: Recognition error rate against babble noise signal-to-noise ratio for three systems, with bootstrap 95 percent confidence bands, on the 144-sentence severity sweep.

almost nothing relative to their translation oracle, the text translation engine applied to perfect transcripts, which scores 55.5 BLEU against the primary reference. The cascade built on Qwen-Audio 3.1 starts at 53.9 BLEU, the cascade built on the streaming specialist at 51.5, and the end-to-end omni path scores highest at 68.1 with the streaming LiveTranslate service at 59.8. Under degradation the ordering and the magnitudes change dramatically. At 0 dB the Qwen-Audio cascade drops 9.9 BLEU points, the streaming-specialist cascade collapses by 35.7 points to 15.8 BLEU, the omni end-to-end path loses 22.5 points, and the streaming translation service loses 9.3 points from a lower starting level.

Two conclusions follow. The first concerns the frontend hypothesis. The translation quality of a cascade under acoustic degradation is governed almost entirely by the robustness of its recognition stage: swapping the frontend from the streaming specialist to the LLM recogniser at 0 dB is worth 28 BLEU points on the same backend, far more than any architectural change downstream. The second qualifies the common expectation that end-to-end systems avoid cascade error propagation and are therefore more robust. At 0 dB the omni end-to-end path loses a third of its BLEU score, more than the well-frontended cascade

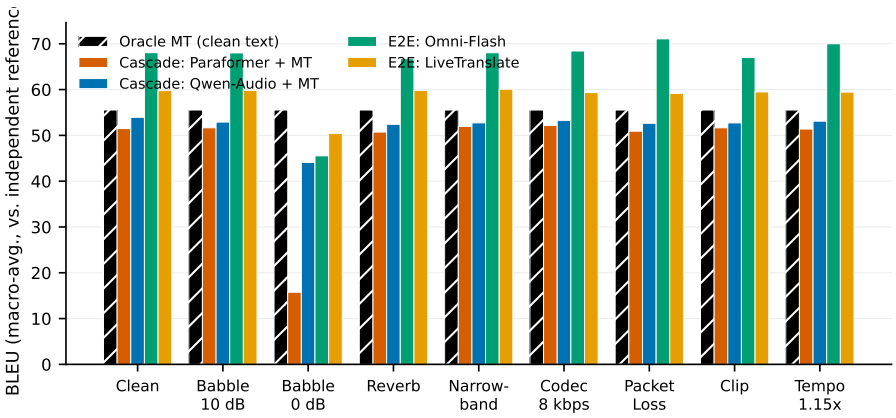


Figure 4: Translation quality of two cascades and two end-to-end speech translation paths across the main degradation grid, scored against the independent LLM reference. The hatched bar is the text-translation oracle on clean transcripts.

loses at 18 percent, and its absolute score falls below that cascade. The streaming LiveTranslate service is the most degradation-resilient of the four, losing 16 percent at 0 dB, which is consistent with the behaviour of its internal frontend recogniser documented in Section 4.7. End-to-end designs do not inherit robustness automatically; they inherit the robustness of whatever acoustic frontend their provider trained internally, which in this measurement is better than the specialised streaming ASR but not automatically better than the best cascade.

The condition-wise structure of Fig. 4 adds three nuances to this picture. First, apart from the 0 dB point the cascade built on the LLM recogniser is essentially flat across the entire grid, moving within one BLEU point of its clean score under every other condition, which mirrors the recognition flatness of Table 2 and confirms that the non-noise degradations are, at deployed severities, engineering noise rather than failure modes for a robust frontend. Second, the streaming translation service shows the same flatness with a single exception: its narrowband score of 60.1 sits marginally above its clean score, and its packet-loss and tempo scores are likewise within half a point of clean, so its 0 dB loss is the only substantive degradation it suffers. Third, the omni end-to-end path exhibits a small but consistent anomaly: its packet-loss and tempo

Table 4: Translation quality by condition: macro-averaged BLEU against the independent reference (primary) and against the oracle cascade output (secondary), with the clean-to-worst relative loss for the 0 dB condition.

System	Clean	Babble 0	Narrowband	Loss at 0 dB	chrF clean
Oracle MT (clean text)	55.5	–	–	–	66.7
Cascade: Paraformer + MT	51.5	15.8	51.9	69%	63.4
Cascade: Qwen-Audio + MT	53.9	44.1	52.7	18%	65.0
E2E: Omni-Flash	68.1	45.5	68.0	33%	73.0
E2E: LiveTranslate	59.8	50.4	60.1	16%	68.5

scores, at 71.1 and 70.0, exceed its clean score of 68.1. Inspection of the outputs suggests that under these mild distortions the model produces slightly more concise translations that happen to align better with the reference style, rather than genuinely better translations; the effect is within the reference-sensitivity margin documented in Section 4.5 and is a further argument for treating small absolute differences between heterogeneous systems with caution.

Decomposing the 0 dB collapse by direction sharpens the frontend story. The specialist cascade loses 26 BLEU points in the Chinese-to-English direction but 46 points in the English-to-Chinese direction, falling to 8.7 BLEU, because its English recognition, at 75.5 percent error, destroys the source representation before the translation stage operates. The LLM-frontend cascade degrades far less and nearly symmetrically, losing 7.0 points in one direction and 15.0 in the other. The streaming translation service loses almost nothing in the Chinese-to-English direction at 0 dB, at 55.5 against 55.9 clean, but 18 points in the other direction, which parallels the language asymmetry of its internal frontend. Direction-symmetric deployment claims should therefore be checked per direction rather than pooled.

4.5. Reference sensitivity of translation scoring

Table 4 also reports scores against the secondary reference, the oracle cascade output. The two references invert the apparent ordering of the systems on

clean audio. Against the independent LLM reference the omni end-to-end path leads at 68.1 BLEU, exceeding even the text-translation oracle at 55.5. Against the oracle reference the same end-to-end path scores 48.5 while the Qwen-Audio cascade scores 89.6, since the cascade and the reference generator are the same model. The inversion quantifies a hazard for any evaluation that scores machine translation against machine-generated references: absolute scores, and even the identity of the apparent best system, depend on which model family generated the reference. Two safeguards make the present conclusions robust to this effect. Degradation losses are computed within a system against a fixed reference, so the percentage losses reported above do not depend on the choice, and the ordering of losses at 0 dB, at 69, 33, 18, and 16 percent for the four paths, is preserved under both references. We recommend that future black-box translation evaluations report at least one reference from a model family disjoint from every system under test, and state the family relationships explicitly.

4.6. Error pass-through from recognition to translation

Fig. 5 relates the BLEU loss of the two cascades to their mean recognition error rate across all twelve operating conditions. The relationship is strikingly linear: the Pearson correlation between recognition error and translation loss is 0.988 for the Qwen-Audio cascade and 0.999 for the specialist cascade over twelve points spanning 0.1 to 59 percent recognition error. The fitted pass-through slopes are 0.47 BLEU points per percentage point of recognition error for the Qwen-Audio cascade and 0.66 for the specialist cascade. The steeper slope for the specialised frontend reflects the composition of its errors under severe noise, which include whole-utterance failures where the translation engine receives little or no usable signal, whereas the LLM recogniser's residual errors under moderate degradation are concentrated in individual words that the translation engine can often route around. For engineering purposes the rule is simple and accurate in this measurement: every percentage point of recognition error costs about half a BLEU point downstream, and under conditions that produce outright recognition failures the cost approaches two thirds of a point. A service

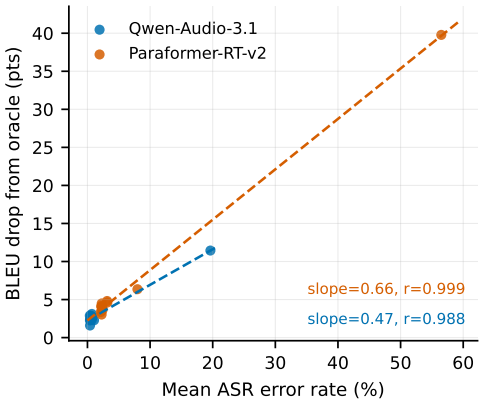


Figure 5: Translation quality loss against mean recognition error rate across twelve operating conditions for two cascades, with fitted linear pass-through.

integrator can therefore estimate the translation consequence of a candidate frontend from recognition error measurements alone, without running the full pipeline.

4.7. The streaming translation frontend as a recogniser

The streaming translation service exposes its internal recognition output, which allows its frontend, a specialised streaming recogniser, to be scored as a sixth system on the severity sweep. Its clean-condition error rate of 0.9 percent is comparable to the other specialised systems, but under noise it behaves markedly better, reaching 18.2 percent at 0 dB, against 25 to 28 percent for the LLM-ASR systems and 58.5 percent for the general-purpose streaming specialist, and it does so at the streaming latency of its class. This is a noteworthy observation for practitioners: the most noise-robust recogniser measured in this study is not directly purchasable as a recognition service but is embedded inside a translation product, which suggests that providers optimise internal streaming frontends for exactly the acoustic conditions where end-to-end translation must survive.

4.8. Sentence-level variability and latency

Fig. 6 shows the sentence-level error distributions for the strongest LLM recogniser and the specialised streaming recogniser on five representative conditions. The distributions are strongly right-skewed: under moderate degradation most sentences remain nearly correct while a tail of sentences fails completely, which is why the confidence intervals in Table 2 are wide relative to their means. Under 0 dB babble the median sentence is still recognised reasonably well by the LLM systems while the upper quartile fails outright, a heterogeneity that aggregate error rates conceal and that matters for applications with per-utterance quality requirements. The skew also explains a statistical property visible throughout Table eftab:main: the confidence intervals of the LLM systems under moderate degradation are wide relative to their means because a small number of complete failures, including the refusal cases, coexist with a large mass of near-perfect transcripts. For a system operator this shape matters more than the mean, since the tail sentences are the ones that generate complaints and downstream errors, and the fraction of sentences above a given error threshold is a more actionable deployment metric than the average. On the 0 dB condition, the fraction of English sentences with error rates above 50 percent is 83 percent for the specialised streaming recogniser against 36 percent for Qwen3-ASR, while in Chinese the same threshold is exceeded by 33 percent of sentences for the specialised system against 3 percent for Qwen-Audio, so the family gap in tail behaviour is considerably larger than the gap in means.

Table 5 reports API round-trip latencies. The text translation stage of the cascade adds about 0.27 seconds per sentence, so a Qwen-Audio cascade completes in about 1.1 seconds on average against 0.8 seconds for its recognition stage alone, whereas the omni end-to-end path averages 2.8 seconds with a 95th percentile above 7 seconds, reflecting the substantially longer generation path of an omni-modal foundation model that produces its translation in a single response. The streaming translation service completes in under a second. Latency therefore reinforces rather than reverses the accuracy picture: the paths that degrade least under noise, the well-frontended cascade and the streaming

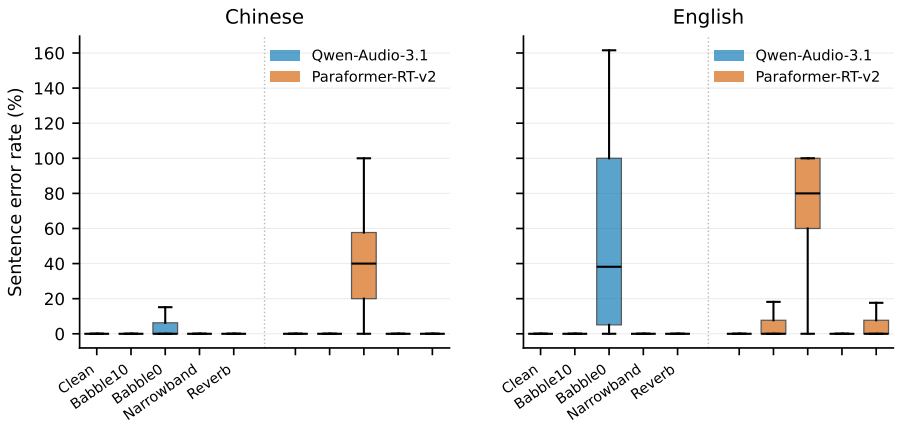


Figure 6: Sentence-level error distributions by condition for the strongest LLM recogniser and the specialised streaming recogniser.

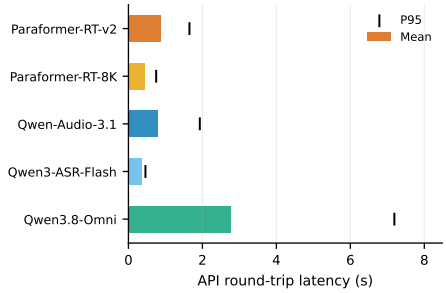


Figure 7: API round-trip latency by system, mean and 95th percentile.

translation service, are also among the fastest, while the slowest path, the omni end-to-end model, is the most fragile under severe noise in this measurement.

5. Discussion

5.1. Guidance for system integrators

The measurements translate directly into selection guidance for practitioners who compose commercial speech services. Above a signal-to-noise ratio of about 5 dB, all five recognition systems deliver sub-percent to low-percent error rates on clearly articulated speech, and the choice among them can be driven

Table 5: API round-trip latency in seconds for the recognition systems and the text translation stage.

System	Mean	P50	P95
Qwen3-ASR-Flash	0.37	0.35	0.46
Paraformer-RT-8K-v2	0.45	0.43	0.75
Qwen-Audio-3.1-Flash	0.80	0.72	1.93
Paraformer-RT-v2	0.87	0.85	1.65
Text translation (qwen-mt-uni)	0.27	0.26	0.39
Qwen3.8-Omni-Flash (E2E)	2.77	2.20	7.19
LiveTranslate (E2E)	0.55	0.52	0.80

by latency, cost, and interface ergonomics rather than accuracy. The latency data in Table 5 then favour the LLM-ASR text-input systems and the specialised streaming recognisers, which all answer in well under a second, over the omni-modal path. Between 0 and 5 dB, the architectural families separate: the degradation gradients in Table 3 show that every dB of noise costs a specialised streaming recogniser roughly four to five times what it costs an LLM-ASR system in Mandarin and about twice as much in English, so in intermittently noisy environments such as public spaces, vehicles, or shared offices, the foundation-model recognisers dominate the accuracy-latency trade-off. Below 0 dB, no system transcribes English reliably, and the appropriate response is architectural: capture improvement, multi-microphone processing, or explicit rejection logic. For telephone-band audio, the surprising result is that no specialist premium exists; the generalist systems are as accurate in-band, and the telephone specialist is markedly worse under reverberation, so its selection can be justified only by its 8 kHz interface requirements and its speed, not by accuracy.

For translation pipelines, the frontend hypothesis dominates all other choices, and the latency data combine with it into a compact decision rule. A cascade assembled from the fastest recognition tier and the text translation engine completes in about 1.1 seconds per sentence at the mean, roughly twice as fast as the omni end-to-end path, while the streaming translation service completes in

about half a second and is therefore the natural choice when latency dominates; when noise robustness dominates, the same streaming service and the LLM-frontend cascade lead the field, so the two objectives point to the same shortlist. Swapping the recognition stage of a cascade from the specialised streaming system to the LLM recogniser is worth 28 BLEU points at 0 dB on an identical translation backend, which exceeds any reasonable expected gain from changing the translation engine itself. The linear pass-through rule quantified in Section 4.6, at roughly half a BLEU point per percentage point of recognition error, allows an integrator to rank candidate frontends with recognition-only measurements, which are cheaper to run and easier to interpret than full-pipeline evaluations, and to forecast translation quality degradation from a measured or contracted recognition error rate. The end-to-end versus cascade decision should rest on measured degradation behaviour rather than architectural intuition: the omni-modal end-to-end path evaluated here is simultaneously the slowest and, under severe noise, the most fragile of the four translation designs, while a dedicated streaming translation service achieves both the smallest relative loss at 0 dB and sub-second latency.

5.2. Why foundation-model recognisers degrade more slowly

The robustness advantage of the LLM-ASR systems echoes the finding of Radford et al. [9] that recognition models trained on very large and weakly curated corpora acquire substantial noise tolerance without explicit multi-condition training. Two of our observations sharpen this picture for API consumers. First, the advantage is strongly condition-dependent: it is large under energetic babble noise and negligible under codec, packet-loss, clipping, and tempo distortions, which barely affect any system. This pattern is consistent with robustness arising from exposure to diverse recording conditions during pre-training rather than from any explicit enhancement front-end. Second, the advantage is language-asymmetric, with Chinese recognition nearly immune at 0 dB while English recognition of the same systems fails at 37 to 50 percent. Providers disclose neither training mixture nor per-language capacity, so the asymmetry

can only be documented, not explained, at the black-box level; the documentation itself is actionable, since it implies that multilingual deployments should be derated per language rather than per provider. It also implies a concrete procurement test: a single-language noise probe, one hundred sentences of the target language at 0 dB babble, is enough to place a candidate service on the robustness spectrum documented here, and repeating the probe per deployment language guards against extrapolating a strong Mandarin result to a weak English result or the reverse. The direction decomposition of Section 4.4 extends the same caution to translation deployments, where the two directions of a bilingual service degraded by strikingly different amounts under identical noise.

5.3. Cascade versus end-to-end revisited

The classical comparison of Bentivogli et al. [12] found end-to-end and cascade systems competitive on clean speech with different error structures. Our measurements extend that comparison to degraded channels in the black-box setting and qualify it in two ways. The end-to-end advantage in absolute quality against an independent reference is real but partly a family artefact, since the omni system that translates directly also shares a generation family with the strongest reference producer, and the advantage inverts under the secondary reference, as Section 4.5 shows. More importantly, robustness is not an architectural property but a property of the acoustic frontend wherever it lives. The end-to-end service with the specialised internal frontend is the most resilient path we measured, while the end-to-end omni path is the most fragile, and the same frontend ordering that explains cascade behaviour also explains end-to-end behaviour. For a consumer, the practical question is not which architecture to prefer but where the provider placed its best acoustic model, and the answer is discoverable only by measurement of the kind this benchmark performs.

5.4. Methodological implications for black-box evaluation

Three methodological observations generalise beyond the specific services tested. The dual-reference protocol of Section 4.5 demonstrates that machine-

generated evaluation references can reorder systems in a way that tracks reference-generator affinity rather than translation quality, and that degradation-relative conclusions are the stable currency of black-box comparison. The synthesis-based corpus design gives black-box studies an exact transcript by construction, which matters because at sub-percent clean error rates, transcription uncertainty in recorded corpora would be comparable to the effects being measured. Finally, the refusal behaviour documented in Section 4.1, where one system returns empty output on a small but systematic subset of severely degraded utterances, argues that black-box benchmarks should report response completeness alongside error rates, since a missing transcript and a wrong transcript have different downstream failure modes.

5.5. A worked example of the pass-through rule

The pass-through rule is most useful when stated as an explicit calculation. Suppose an integrator operates a cascade built on the LLM recogniser in an environment that occasionally degrades to 10 dB, at which the recogniser errs at 0.9 percent, and contracts a translation stage equivalent to the one measured here. The rule predicts a translation loss of roughly 0.47 times 0.9, that is 0.4 BLEU points, which is below the clean-condition agreement between the two reference systems and therefore negligible. If the same deployment used the specialised streaming recogniser, whose 10 dB error rate is 2.8 percent, the predicted loss is 1.9 BLEU points, still modest, but the margin to the 0 dB regime, where the rule predicts 27 to 38 points of loss for the two frontends respectively, is thin. The rule therefore both quantifies the mild regime, where no architectural action is needed, and identifies the severity threshold beyond which the frontend choice becomes the dominant design decision. The same arithmetic applied to the measured error rates of a candidate system at the actual operating signal-to-noise distribution of a deployment yields an expected translation quality distribution without running the translation stage at all.

5.6. Implications for evaluation practice and future work

Beyond the specific findings, the benchmark design carries lessons for future evaluations of commercial speech intelligence. A benchmark intended to track black-box services over time should log every raw response rather than only aggregate scores, because behavioural changes such as the refusal pattern documented here appear as distributional shifts before they appear in means. It should treat reference construction as a first-class experimental variable, since Section 4.5 shows that the reference generator can reorder systems. It should separate degradation gradients, which generalise across severity, from point estimates, which do not, and it should report response completeness alongside accuracy. The present study points to four directions for follow-up work: replication across providers to separate family effects from provider effects, extension to spontaneous and accented speech, sweep-based characterisation of the currently fixed-severity conditions, and human-reference scoring of the translation stage to anchor the machine-reference comparisons. As foundation model services begin to expose streaming speech interfaces with embedded recognition, the measurement approach demonstrated here, treating each service as an opaque function from degraded audio to text and scoring the complete pipeline end to end, appears to be the only viable methodology for the comparative robustness questions that practitioners actually face.

6. Limitations

The benchmark uses synthetic speech from four neural voices. Synthetic speech lacks the disfluencies, accents, emotional variation, and channel noise of spontaneous human recordings, and although the exact-transcript property is decisive for measurement at sub-percent error floors, clean error rates measured on synthetic speech should not be read as population estimates for human speech; the degradation deltas, which are the object of this study, are less sensitive to this concern because they are within-system contrasts. All systems come from a single provider, so architectural family effects and provider effects are

confounded; the family-level regularities reported here are consistent with published training-scale explanations [9], but cross-provider replication is required before they can be attributed to architecture rather than to one provider's engineering. The babble noise is itself synthesised from disjoint neural voices, which makes it speech-shaped and fully reproducible but smoother than real multi-talker cocktail-party interference, which includes semantic competition and stronger nonstationarity. Severity levels for the non-noise degradations were fixed at deployed-but-aggressive values rather than swept, so only the noise family is characterised by a gradient. Translation quality is measured with BLEU and chrF against machine-generated references, with the family-affinity bias that Section 4.5 quantifies; human-reference scoring would strengthen absolute claims. Finally, commercial APIs are moving targets: the measurements document the October 2026 versions of the listed services, and one of the systems could not complete its full condition grid before its free-tier quota was exhausted, which is itself a practically relevant datum for anyone planning a similar measurement on complimentary quotas.

7. Conclusions

This paper introduced CascadeBench, a black-box benchmark that measures how acoustic channel degradations propagate through commercial speech recognition and translation services, and reported a complete measurement over a controlled bilingual corpus of 480 sentences, eight degradation conditions, and a five-level noise severity sweep, covering five recognition systems from three architectural families, two cascade translation paths, and two end-to-end speech translation paths, all accessed through their public interfaces on free service quotas.

The measurements support five conclusions. First, the architectural families of commercial recognisers differ sharply and systematically in noise robustness: LLM-based recognisers degrade at 0.34 to 0.44 percentage points per dB in Mandarin against 1.70 for a specialised streaming recogniser, and the family

gap exceeds a factor of two in pooled error rate at 0 dB. Second, robustness is condition- and language-specific rather than uniform: narrowbanding, coding, packet loss, clipping, and tempo changes at deployed severities are individually negligible for all systems, while 0 dB babble nearly destroys English recognition for every system and leaves the best Mandarin recognition almost intact. Third, the quality of a speech translation cascade under degradation is governed by its recognition frontend, and the relationship is linear and predictable, with a pass-through of 0.47 to 0.66 BLEU points per percentage point of recognition error at a correlation above 0.98 across twelve conditions. Fourth, end-to-end speech translation is not automatically more robust than a cascade: the most resilient translation path measured is a dedicated streaming end-to-end service, while the omni-modal end-to-end path is both the slowest and, under severe noise, the most fragile. Fifth, black-box evaluation carries specific methodological obligations that this benchmark operationalises, including dual-reference scoring to control machine-reference affinity, exact-transcript corpora to enable sub-percent measurement, and the reporting of response completeness alongside error rates.

The benchmark recipe, including the corpus, the seeded degradation pipeline, all model responses, and the analysis code, is released so that the measurement can be re-executed as providers update their services. The released artefacts are versioned alongside the service identifiers and access dates of every system, so that future re-runs can attribute score movements either to provider-side model updates or to changes in the benchmark itself, a distinction that periodic evaluation of moving targets requires. Together with the zero-cost execution protocol described in Section 3, the release is intended to make robustness re-measurement a routine maintenance activity for speech deployments rather than a one-off research exercise. As speech intelligence continues to migrate into provider-operated black-box systems, controlled measurement of this kind, repeated over time and across providers, offers the community a practical substitute for the training-set access that classical robustness research assumed, and offers integrators the quantitative basis that architectural intuition alone

cannot supply.

Appendix A. Corpus composition

Table A.6 summarises the corpus structure and lists one sample sentence per stratum. The four domains are constructed to differ in register, vocabulary, and numerical density. The broadcast-news domain uses formal written register with institutional vocabulary, the conversation domain uses colloquial everyday phrasing, the technical domain uses scientific and engineering terminology, and the numerals-and-entities domain concentrates dates, times, quantities, prices, addresses, codes, and named entities, with all numbers written out in words as described in Section 3. Length strata are defined per language as short, medium, and long as given in Section 3, and every domain-by-length cell contains twenty sentences per language, giving 240 sentences per language and 480 in total. The severity sweep subset of 144 sentences is sampled evenly across all cells, six sentences per cell per language.

Table A.6: Corpus structure with one sample sentence per domain.

Domain	Sample sentence
News (zh)	
News (en)	The central bank raised interest rates by twenty five basis points to curb inflation.
Conversation (zh)	
Conversation (en)	I was thinking we could drive to the beach this weekend if the weather holds.
Technical (zh)	
Technical (en)	Reverberation arises when sound waves reflect repeatedly off room surfaces.
Numerals (zh)	
Numerals (en)	The high speed train from Beijing to Shanghai takes about four and a half hours.

Appendix B. Qualitative error analysis

Table B.7 collects representative model responses that illustrate the failure patterns behind the aggregate numbers in Section 4.

The refusal behaviour of Section 4.1 is shown in the first row: under 0 dB babble the Qwen-Audio recogniser returns an empty transcript for a substantial English sentence, which the benchmark scores as a complete failure. The second row shows the opposite regime: the same system, on a long English news sentence at the same severity, produces a verbatim transcript, which underlines how bimodal the sentence-level distributions of Fig. 6 are. Row three shows the dominant error mode of the specialised streaming recogniser under severe noise, where most of the sentence is lost and only a short fragment survives. Row four shows a typical near-miss of the LLM recogniser in Chinese, where a single content word, density, is replaced by another word of similar phonetic shape, speed, without any structural damage. Row five shows the collapse of the telephone-band specialist under reverberation, which occasionally produces short disconnected fragments rather than transcripts. Row six shows an error that the recognition stage never sees but the translation consumer experiences: under 0 dB noise the omni end-to-end path substitutes a plausible but wrong entity, rendering the sports centre of the source as a grand theatre, a hallucination that a string-matching metric penalises only lightly but that is catastrophic for factual fidelity. Row seven shows the narrowband immunity of the LLM recogniser, which returns a verbatim transcript of telephone-band audio. Together the examples support the reading of Section 5: the systems differ less in whether they fail than in how, and the failure modes, from verbatim robustness to silent entity substitution to outright refusal, have distinct downstream consequences.

Appendix C. Cross-system text normalisation

Comparing transcripts across commercial recognisers requires a normalisation protocol that neutralises provider-specific inverse text normalisation. The protocol applies the following rules identically to references and hypotheses, in

Table B.7: Representative system responses under degradation.

Condition	System	Reference and response
0 dB, en	Qwen-Audio-3.1	Reference: Police arrested two suspects on Tuesday. Response: empty transcript.
0 dB, en	Qwen-Audio-3.1	Reference: The transport minister told reporters on Friday that the government would invest heavily in electric vehicle charging infrastructure. Response: verbatim transcript.
0 dB, en	Paraformer-RT-v2	Reference: The battery lasts twenty hours. Response: About.
0 dB, zh	Qwen-Audio-3.1	Reference: Response:
Reverb, zh	Paraformer-RT-8K	Reference: Response:
0 dB, E2E	Omni-Flash	Source: Translation: The newly built Grand Theatre can accommodate 60,000 audience members.
Narrow, en	Qwen-Audio-3.1	Reference: This coffee shop is cozy. Response: verbatim transcript.

order. Unicode compatibility folding and case folding are applied first, and punctuation is removed. Numeric expressions are then canonicalised to digit form on both sides. For Chinese, the rules handle cardinal numbers with the standard multipliers, digit-by-digit readings such as room and code numbers where each digit is read separately, decimal fractions written with the point particle, time-of-day expressions in which the hour and minute parts are converted separately, percentage markers, which are deleted on both sides, and scale multipliers, where a digit followed by wan, yi, or their combinations is expanded to the full multiplied-out form, so that twenty-six million written as 2600 in one system's output and as 26000000 in another's both canonicalise

identically. For English, the rules handle cardinal numbers including compound hyphenated forms and the and connector, time and digit-by-digit readings such as twelve zero eight and nine forty, which canonicalise to 1208 and 940, decimal fractions expressed with point, scale multipliers from thousand to trillion, and ordinal expressions of both word and digit form, which are masked to a common placeholder since their spoken-to-written realisations differ across systems in ways that carry no acoustic information. The rule set was validated by inspecting all high-error sentences of the clean condition until no residual mismatches attributable to normalisation remained, and the clean-condition error rates of Table 2, which reach 0.14 percent for the strongest system, bound the residual protocol noise from above.

Appendix D. Complete pairwise significance results

Table D.8 reports all pairwise Wilcoxon signed-rank comparisons between recognition systems that are significant at the Holm-corrected level, grouped by condition and language. Every reported pair is significant after correction across the family of comparisons within each condition, which is why the table lists only significant results. The consistency of the ordering is itself informative: within every condition and language, the two LLM-ASR systems and the omni model occupy the leading positions and the specialised streaming recogniser the trailing position, with the single systematic exception that Qwen3-ASR leads Qwen-Audio 3.1 in English under reverberation, clipping, and 0 dB noise, where the refusal behaviour of the latter penalises it.

Appendix E. Numeric results of the severity sweep

Table E.9 reports the complete error rates of the severity sweep with confidence intervals, and Table E.10 reports the internal frontend recogniser of the streaming translation service on all twelve conditions. Two details of the sweep data deserve note. The confidence intervals at 0 dB are wide for every system,

Table D.8: Significant Holm-corrected pairwise Wilcoxon comparisons by condition and language. Cells give the mean error rates in percent of the two systems compared.

Condition, language	Comparison	Means (%)	p
0 dB, zh	Qwen-Audio vs Paraformer-RT	6.3 vs 41.6	< 0.001
0 dB, zh	Qwen3-ASR vs Paraformer-RT	13.1 vs 41.6	< 0.001
0 dB, zh	Qwen-Audio vs Qwen3-ASR	6.3 vs 13.1	< 0.001
0 dB, zh	Qwen-Audio vs Omni	8.5 vs 20.2	< 0.001
0 dB, zh	Paraformer-RT vs Omni	42.4 vs 20.2	< 0.001
0 dB, en	Qwen-Audio vs Paraformer-RT	50.2 vs 75.5	< 0.001
0 dB, en	Qwen3-ASR vs Paraformer-RT	37.1 vs 75.5	< 0.001
0 dB, en	Qwen-Audio vs Qwen3-ASR	50.2 vs 37.1	< 0.001
0 dB, en	Paraformer-RT vs Omni	76.1 vs 43.3	< 0.001
Reverb, zh	Qwen-Audio vs Paraformer-RT	0.2 vs 1.3	< 0.001
Reverb, en	Qwen3-ASR vs Paraformer-RT	0.5 vs 5.1	< 0.001
Reverb, en	Qwen-Audio vs Qwen3-ASR	1.2 vs 0.5	0.011
Narrow, zh	Qwen-Audio vs Paraformer-RT	0.2 vs 0.7	< 0.001
Narrow, en	Qwen3-ASR vs Paraformer-RT	0.5 vs 3.6	< 0.001
Narrow, en	Qwen-Audio vs Qwen3-ASR	1.0 vs 0.5	0.015
Clip, en	Qwen3-ASR vs Paraformer-RT	0.4 vs 4.4	< 0.001
Clip, en	Qwen-Audio vs Qwen3-ASR	1.6 vs 0.4	< 0.001

spanning roughly ten percentage points, which reflects the bimodal sentence-level behaviour documented in Section 4.8. The crossover between the families occurs between 5 and 0 dB for both languages, and at 5 dB the specialised system already shows its first significant departure from its clean score while the LLM-ASR systems remain within 0.3 points of theirs in Mandarin. The internal frontend of the streaming translation service records a zero lower confidence bound at 5 dB, its strongest operating point, and degrades less at 0 dB than any directly purchasable recognition service measured in this study.

Table E.9: Recognition error rates in percent on the severity sweep subset, 144 sentences, with bootstrap 95 percent confidence intervals.

System	Lang	20 dB	15 dB	10 dB	5 dB	0 dB
Paraformer-RT-v2	zh	0.8	0.8	0.8	2.5	42.4
		(0.3,1.3)	(0.3,1.4)	(0.3,1.4)	(1.1,4.2)	(36.0,49.5)
	en	3.8	3.7	3.9	13.4	76.1
		(1.9,6.0)	(1.9,5.9)	(2.1,6.0)	(9.3,17.6)	(71.2,80.8)
Qwen-Audio-3.1	zh	0.1	0.1	0.1	0.1	8.5
		(0.0,0.2)	(0.0,0.2)	(0.0,0.2)	(0.0,0.2)	(4.3,13.8)
	en	0.9	1.3	1.5	2.0	39.7
		(0.1,2.3)	(0.3,2.6)	(0.4,2.9)	(0.8,3.6)	(30.3,49.9)
Qwen3-ASR-Flash	zh	0.1	0.2	0.1	0.3	11.2
		(0.0,0.4)	(0.0,0.5)	(0.0,0.4)	(0.0,0.7)	(7.1,15.7)
	en	0.9	0.9	0.7	1.9	34.1
		(0.0,2.2)	(0.1,2.2)	(0.0,1.9)	(0.6,3.8)	(25.8,43.4)

Data and code availability

The full benchmark recipe, including the sentence corpus, the perturbation implementations with fixed random seeds, all per-sentence model transcripts and translations, and the analysis scripts, will be made publicly available at a repository link provided upon acceptance. The data can also be reconstructed end to end from the documentation in this paper using free-tier API quotas.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table E.10: Error rate in percent of the streaming translation service internal frontend recogniser on all twelve conditions.

Condition	Clean	20 dB	15 dB	10 dB	5 dB	0 dB
Error (%)	0.9	0.8	0.2	0.9	0.0	18.2
95% CI	(0.3,1.7)	(0.3,1.6)	(0.0,0.6)	(0.3,1.6)	(0.0,0.0)	(13.9,23.2)
Condition	Reverb	Narrow	Codec	Loss	Clip	Tempo
Error (%)	0.8	0.8	0.7	0.8	0.8	1.0
95% CI	(0.3,1.5)	(0.3,1.6)	(0.2,1.4)	(0.3,1.5)	(0.3,1.6)	(0.4,1.8)

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